What 394,300 job postings reveal about pay, skills and hiring

Marc Abi Zeid Daou · Data source: zalizedata/tech-job-postings-salary-dataset (CC BY-NC 4.0) · Snapshot of open roles, August 2026

Summary. This project analyzes 394,300 job postings published by 6,295 companies in 65 countries, of which 64,574 advertise a yearly salary in US dollars. The typical (median) advertised salary is **$160k** (95% CI $155k–$165k). Pay rises 2.7× from entry-level to principal/staff titles. Remote postings advertise 10% more than on-site postings, but comparing similar jobs within the same company the premium falls to **4.7%** (95% CI 2.5% to 6.9%). A regularized linear model predicts advertised salaries with R^2 = **0.77** on held-out postings (typical error 11%); job title and employer carry most of the signal, while individual skills shift pay by at most 4%.

1. Questions

(1) What do these jobs advertise, and which job types pay the most? (2) How much does seniority matter? (3) Do remote jobs pay more once comparable jobs are compared? (4) Do specific skills raise pay? (5) How accurately can a machine-learning model predict an advertised salary?

2. Data

Postings were collected from public company career pages hosted on applicant-tracking systems (SmartRecruiters, Greenhouse, Ashby, Lever, Workable). The data covers every type of role: tech jobs (software, data, product, security) are 21% of postings with an identified job type. Only 27% of postings state any salary, and disclosure follows the law: 56% of US postings do, versus 3% in Germany and 1% in France. As a result, 82% of usable salaries are US jobs, and figures describe these postings rather than national pay levels.

Salary information	Postings
Yearly USD salary (used)	64,574
No salary listed	287,149
Other currency or pay period	42,029
Range wider than 4×	303
Midpoint outside $20k–$1M	200
Invalid range	45

Salaries are the midpoint of the advertised range. Only yearly USD ranges are used, so no currency or hours-per-year conversion is assumed; midpoints outside $20k–$1M (typically hourly rates mislabeled as yearly) and ranges wider than 4× are excluded. **Job type** (19 groups) and **seniority** (5 levels) are derived from title keywords with ordered rules that include French, German, Spanish and Dutch terms; 86% of postings (96% of salaried ones) match a job type. **Validation.** Most postings also carry a department chosen by the employer (e.g. "Engineering", "Sales"), which is independent of the title rules.

Mapping clear department names to groups and allowing plausible overlaps (a data job in an engineering department counts as a match), the title-based job type agrees with the employer's department for **85%** of 117,772 postings from 4,706 companies (95% Wilson CI 85.0%–85.4%), from 93% for Product & Design down to 60% for Skilled Trades & Field. This check also exposed a rule error: titles such as "Sales Engineer" and "Technical Support Engineer" had been classified as engineering because of the word "engineer"; they are now classified as sales and support jobs. Cross-functional job types (operations, consulting, management) appear in every kind of department, so the department cannot confirm them; they are not scored.

3. Methods

Uncertainty. Postings from one company share pay bands and are often copied across cities, so they are not independent. Confidence intervals for medians therefore use a *company cluster bootstrap*: 1,000 replicates, each resampling whole companies with replacement; intervals are the 2.5th–97.5th percentiles.

Remote pay. Ordinary least squares regressions of log salary on a remote indicator, adding controls step by step: job type and seniority; country (top 10) and employment type; and finally company fixed effects (the within-company estimator). Standard errors are clustered by company. Duplicate postings (same company, title and salary range) are collapsed, and unclassified titles are excluded (n = 52,749).

Salary prediction. The target is log salary. Inputs are title text (TF-IDF unigrams and bigrams), company, industry, department, city, country, employment type, remote flag, skills and title-derived labels. After collapsing duplicates (64,574 → 54,859), 20% of postings are held out. Four candidates are compared by 5-fold cross-validation on the remaining 80%; the winner is scored once on the held-out set and again on a split in which entire companies are unseen. An 80% prediction interval is formed from out-of-fold residuals (split-conformal style) and its coverage is checked on the test set.

4. Results

4.1 Pay by job type

The highest-paying job types are Software Engineering ($195k), Data & AI ($191k), Product & Design ($190k), Security ($188k); their intervals overlap, so their exact order is not statistically distinguishable. At the other end are Students & Apprentices ($62k), Education ($74k), Retail & Hospitality ($80k). Small groups have wide intervals (e.g. Healthcare $103k–$195k), a reminder that a median from a few hundred postings is far less certain than one from ten thousand.

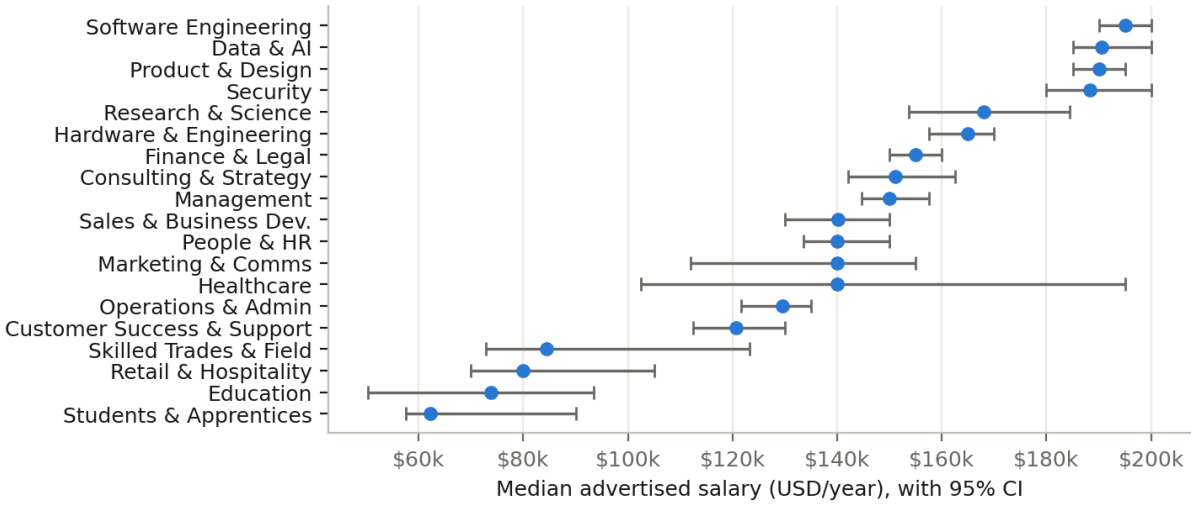


Figure 1. Median advertised salary by job type, with 95% company cluster bootstrap confidence intervals.

4.2 Seniority

Median advertised pay rises from $83k for entry-level titles to $223k for principal/staff titles, a 2.7× difference (95% CI 2.3×–3.3×). Director/executive titles ($204k) sit slightly below principal/staff, plausibly because "director" covers a wide range of organization sizes and functions.

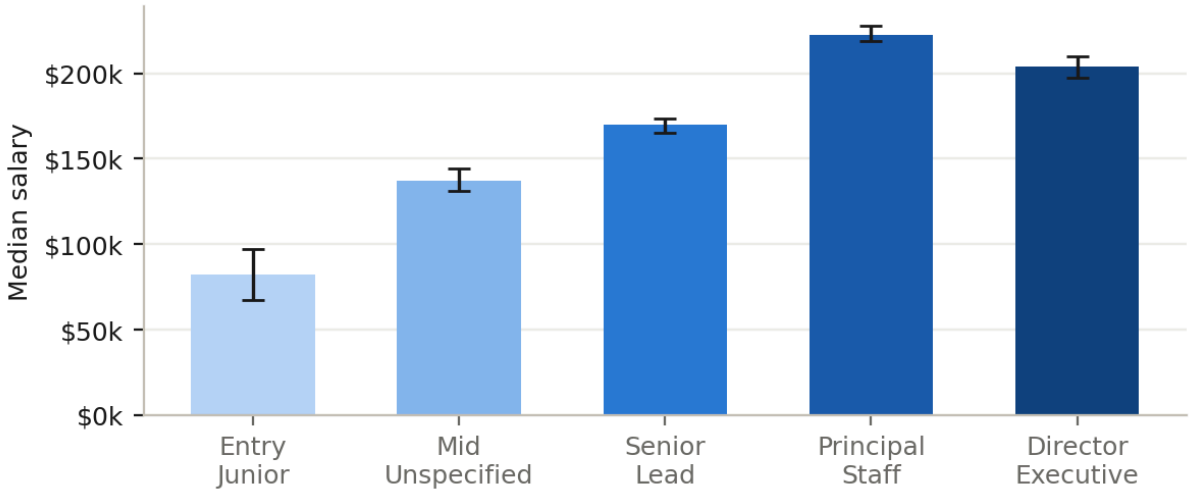


Figure 2. Median salary by title-inferred seniority (whiskers: 95% CI).

4.3 Remote vs. on-site

Remote postings advertise +9.8% more than on-site postings. Part of this gap is composition: remote postings are more often senior and in higher-paying job types. Controlling for job type and seniority reduces it to +5.3%; comparing jobs within the same company leaves a smaller but statistically significant premium of +4.7% (95% CI +2.5% to +6.9%, p = 3e-05).

Comparison	Remote difference	95% CI	p-value
Raw difference	+9.8%	+3.5% to +16.4%	0.002
+ job type & seniority	+5.3%	+0.5% to +10.2%	0.029
+ country & employment type	+6.2%	+1.3% to +11.4%	0.013
+ same company (fixed effects)	+4.7%	+2.5% to +6.9%	2.7e-05

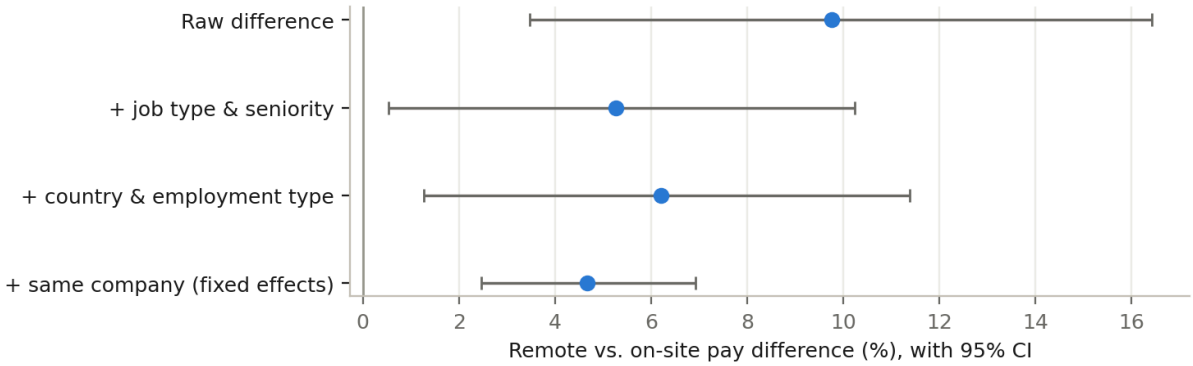


Figure 3. Estimated remote pay difference as controls are added (company-clustered 95% CIs).

4.4 Skills

Postings that list specialized skills advertise high salaries (for example, machine-learning frameworks appear in jobs with medians above $200k). But in the prediction model, which holds title, company and location fixed, no skill listed in at least 300 postings changes predicted pay by more than 3.5%. Skills mostly signal *which* job and employer a posting belongs to, rather than adding pay on their own. These are associations, not causal effects.

4.5 Predicting salaries

Model	CV R^2	Test R^2	Test MAE
Median baseline	-0.004	-0.003	$55,869
Ridge: title + location only	0.586	0.611	$31,437
Gradient boosting (target-encoded)	0.645	0.673	$28,767
Ridge: full feature set	0.737	0.771	$24,047

The selected ridge regression explains 77% of the variance in advertised salaries on held-out postings (R^2 0.771; 0.806 on the log scale), with a mean absolute error of $24,047 and a median absolute error of 11.1%. Its 80% prediction interval contained 80.9% of held-out salaries, so the stated uncertainty is well calibrated. Permutation importance attributes most of the accuracy to the title (R^2 drop 0.58) and the company (0.40); skills contribute 0.015. For companies absent from training, R^2 falls to 0.50: much of the predictive power comes from knowing the employer's pay level. A linear model outperformed gradient boosting here, plausibly because the signal sits in thousands of sparse title words and company identities.

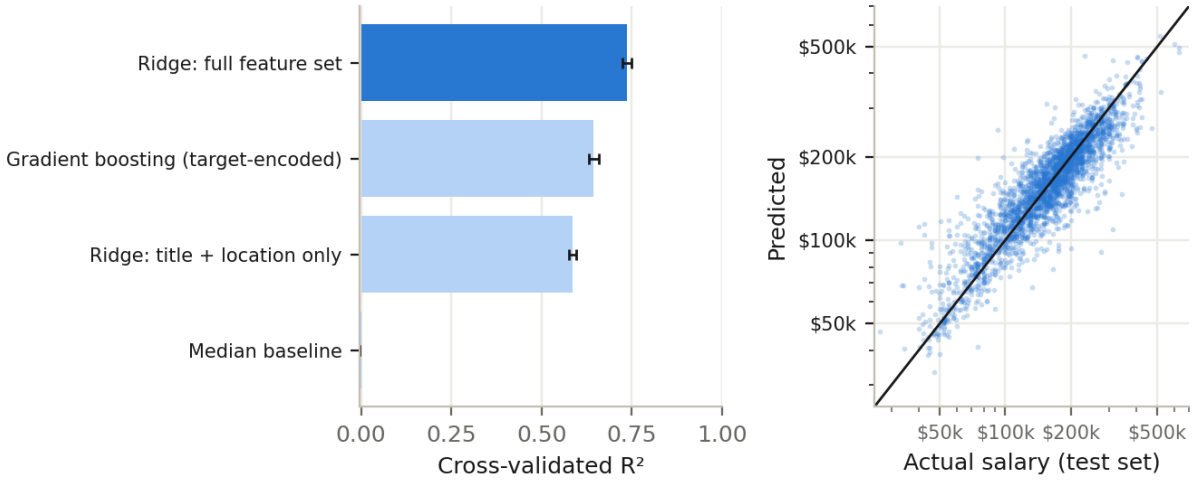


Figure 4. Left: cross-validated R^2 of the candidate models (selected model in dark blue). Right: predicted vs. actual salary for 3,000 held-out postings (log scales).

5. Limitations

Advertised ranges are not realized pay. Salaried postings are concentrated in the US because of disclosure laws, and the companies are those using these hiring platforms, so results describe this population of postings rather than the labor market as a whole. The data is a single snapshot, so it shows how long roles stay open rather than hiring trends. Job type and seniority are rule-based approximations. Regression and skill results are associations; remaining differences between remote and on-site roles (e.g. responsibilities not visible in the title) could still explain part of the premium.

6. Reproducibility

All code, the cleaned data and the trained model are public at github.com/marcazdaou/Job-Market-Analysis. The pipeline is: `pipeline/collector.py` (cleaning) → `models/train.py` (model) → `analysis/salary_statistics.py` (inference) → `analysis/label_validation.py` (job-type check) → `report/build_report.py` (this report). Automated tests check data integrity, model calibration and the dashboard on every push.